

## BACHELOR, Australia

WMO: 941250

Lat: 13.0544S

Lon: 131.0253E

Elev: 344

StdP: 14.51

Time Zone: 9.50 (AUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 6             | 56.7       | 59.5 | 26.1                          | 20.3 | 80.2 | 30.0 | 24.4 | 75.8 | 18.5                  | 79.9 | 17.1 | 78.8 | 2.0                   | 140  | 0.331 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 10            | 21.9                   | 97.9            | 71.7 | 96.3 | 72.2 | 95.0 | 73.0 | 81.6                | 87.3 | 80.7 | 87.1 | 79.9 | 86.4 | 7.0                  | 100  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.3                            | 159.1 | 83.5 | 79.2 | 153.7 | 82.6 | 78.4 | 149.5 | 82.0 | 45.6          | 87.2 | 44.6 | 87.1 | 43.9 | 86.7 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 15.6              | 13.8 | 12.1 | DB                         | 51.6 | 101.3              | 3.2 | 1.9                                                | 49.3 | 102.7      | 47.4 | 103.8      | 45.6 | 104.9      | 43.2 | 106.2 |
|                   |      |      | WB                         | 44.6 | 82.5               | 1.8 | 1.5                                                | 43.3 | 83.5       | 42.2 | 84.4       | 41.2 | 85.2       | 39.9 | 86.3  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 80.9   | 82.1  | 82.1  | 82.4  | 81.6  | 78.5  | 75.6  | 75.6  | 77.1  | 82.4  | 85.0  | 84.8  | 83.5  |
|                                                                               | DBStd       | 4.30   | 2.57  | 2.43  | 2.13  | 2.55  | 3.54  | 4.12  | 3.34  | 3.10  | 2.73  | 2.24  | 2.42  | 2.57  |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | CDD50       | 11272  | 996   | 898   | 1004  | 949   | 883   | 769   | 794   | 841   | 971   | 1086  | 1043  | 1038  |
|                                                                               | CDD65       | 5797   | 531   | 478   | 539   | 499   | 418   | 319   | 329   | 376   | 521   | 621   | 593   | 573   |
|                                                                               | CDH74       | 61054  | 5124  | 4567  | 5318  | 5236  | 4201  | 3070  | 3386  | 4160  | 6042  | 7259  | 6682  | 6009  |
| CDH80                                                                         | 27709       | 1876   | 1688  | 2172  | 2456  | 1895  | 1254  | 1459  | 2015  | 3270  | 3813  | 3258  | 2553  |       |
| Wind                                                                          | WSAvg       | 4.6    | 4.7   | 4.6   | 4.0   | 4.1   | 4.9   | 6.1   | 5.3   | 4.8   | 4.3   | 4.1   | 3.7   | 4.1   |
| Precipitation                                                                 | PrecAvg     | 58.0   | 13.3  | 12.0  | 9.6   | 3.4   | 0.9   | 0.0   | 0.1   | 0.1   | 0.5   | 2.2   | 6.0   | 9.5   |
|                                                                               | PrecMax     | 81.6   | 26.3  | 22.9  | 25.7  | 19.1  | 4.7   | 0.1   | 0.6   | 0.9   | 1.9   | 5.7   | 10.4  | 19.2  |
|                                                                               | PrecMin     | 33.6   | 3.2   | 5.4   | 2.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 2.6   | 2.0   |
|                                                                               | PrecStd     | 10.4   | 5.7   | 5.0   | 6.2   | 4.4   | 1.4   | 0.0   | 0.2   | 0.2   | 0.5   | 1.7   | 2.1   | 5.7   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 94.7  | 94.5  | 94.7  | 95.5  | 94.3  | 92.5  | 92.5  | 94.4  | 98.4  | 100.8 | 99.4  | 97.0  |
|                                                                               |             | MCWB   | 78.7  | 77.9  | 76.5  | 73.2  | 70.8  | 69.5  | 67.9  | 67.5  | 68.6  | 70.5  | 73.6  | 77.2  |
|                                                                               | 2%          | DB     | 92.5  | 92.9  | 93.2  | 93.6  | 92.3  | 90.5  | 90.7  | 92.5  | 96.6  | 98.3  | 97.0  | 94.8  |
|                                                                               |             | MCWB   | 78.8  | 78.1  | 76.2  | 73.3  | 70.6  | 68.8  | 66.9  | 66.8  | 68.3  | 70.9  | 74.6  | 77.5  |
|                                                                               | 5%          | DB     | 91.0  | 91.3  | 91.9  | 92.3  | 90.7  | 88.8  | 89.2  | 91.0  | 95.2  | 96.5  | 95.1  | 93.1  |
|                                                                               |             | MCWB   | 78.8  | 78.3  | 76.3  | 73.2  | 70.5  | 67.8  | 66.0  | 66.1  | 68.7  | 71.4  | 75.5  | 77.7  |
|                                                                               | 10%         | DB     | 89.0  | 89.3  | 90.3  | 90.9  | 89.1  | 87.1  | 87.6  | 89.5  | 93.6  | 94.7  | 93.3  | 91.2  |
|                                                                               |             | MCWB   | 78.6  | 78.2  | 76.8  | 73.3  | 70.2  | 66.7  | 65.2  | 65.3  | 69.0  | 72.3  | 76.1  | 78.1  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 82.5  | 82.3  | 81.9  | 80.3  | 77.7  | 75.5  | 73.2  | 74.7  | 78.5  | 80.9  | 82.5  | 82.1  |
|                                                                               |             | MCDB   | 87.6  | 87.3  | 85.8  | 85.5  | 85.3  | 82.3  | 82.5  | 83.8  | 86.9  | 88.4  | 89.5  | 87.8  |
|                                                                               | 2%          | WB     | 81.0  | 80.9  | 80.6  | 78.9  | 76.4  | 73.4  | 71.8  | 72.9  | 76.5  | 78.9  | 80.4  | 80.7  |
|                                                                               |             | MCDB   | 87.1  | 87.0  | 86.4  | 84.9  | 84.2  | 82.8  | 81.8  | 83.1  | 84.6  | 86.0  | 87.6  | 87.4  |
|                                                                               | 5%          | WB     | 80.1  | 80.0  | 79.7  | 77.9  | 75.5  | 71.8  | 70.3  | 71.7  | 75.4  | 77.7  | 79.2  | 79.8  |
|                                                                               |             | MCDB   | 86.3  | 86.2  | 85.6  | 84.4  | 83.1  | 81.4  | 81.3  | 81.7  | 83.5  | 85.1  | 86.5  | 86.9  |
|                                                                               | 10%         | WB     | 79.3  | 79.2  | 78.9  | 77.1  | 74.3  | 70.2  | 68.9  | 70.4  | 74.4  | 76.8  | 78.3  | 79.1  |
|                                                                               |             | MCDB   | 85.4  | 85.2  | 85.0  | 83.9  | 81.9  | 81.8  | 80.5  | 80.5  | 83.2  | 84.5  | 85.8  | 86.3  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 13.4  | 13.6  | 15.4  | 19.4  | 22.2  | 23.2  | 25.0  | 26.0  | 25.0  | 21.9  | 18.9  | 15.5  |
|                                                                               |             | MCDBR  | 16.9  | 17.5  | 18.8  | 21.5  | 23.4  | 24.1  | 25.9  | 26.5  | 26.0  | 24.2  | 21.1  | 18.9  |
|                                                                               |             | MCWBR  | 5.5   | 5.8   | 5.7   | 6.1   | 7.0   | 7.6   | 8.0   | 7.7   | 7.0   | 6.1   | 5.2   | 5.5   |
|                                                                               | 5% WB       | MCDBR  | 14.5  | 14.8  | 15.1  | 17.3  | 19.0  | 21.2  | 23.9  | 23.6  | 22.9  | 20.7  | 18.6  | 16.2  |
|                                                                               |             | MCWBR  | 5.8   | 6.0   | 5.9   | 5.5   | 6.1   | 7.1   | 7.7   | 6.8   | 6.0   | 5.5   | 5.9   | 5.8   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.418 | 0.408 | 0.395 | 0.372 | 0.369 | 0.347 | 0.343 | 0.359 | 0.426 | 0.449 | 0.453 | 0.420 |
|                                                                               | taud        |        | 2.470 | 2.495 | 2.528 | 2.564 | 2.516 | 2.525 | 2.530 | 2.472 | 2.315 | 2.281 | 2.306 | 2.445 |
|                                                                               | Ebn at Noon |        | 294   | 295   | 293   | 289   | 280   | 281   | 285   | 289   | 279   | 279   | 282   | 293   |
|                                                                               | Edh at Noon |        | 38    | 36    | 35    | 31    | 31    | 30    | 30    | 34    | 42    | 45    | 44    | 38    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 1693  | 1749  | 1802  | 1853  | 1714  | 1647  | 1726  | 1925  | 2074  | 2139  | 2037  | 1840  |
|                                                                               | RadStd      |        | 168   | 264   | 217   | 140   | 92    | 62    | 44    | 59    | 94    | 95    | 86    | 165   |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (8 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                        | N/A   | N/A     | N/A    | +0.43   | N/A   | +0.13 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page